

OFFICIAL TECHNICAL WHITEPAPER

CLOXEL AI

Autonomous Multi-Modal AI Video Generation & YouTube Auto-Publishing Infrastructure

DOCUMENT VERSION

v2.4.0 (Production)

DATE OF AUDIT

September 2026

CORE ENGINE

Google Gemini 2.5/2.0 +
FFmpeg



Executive Abstract

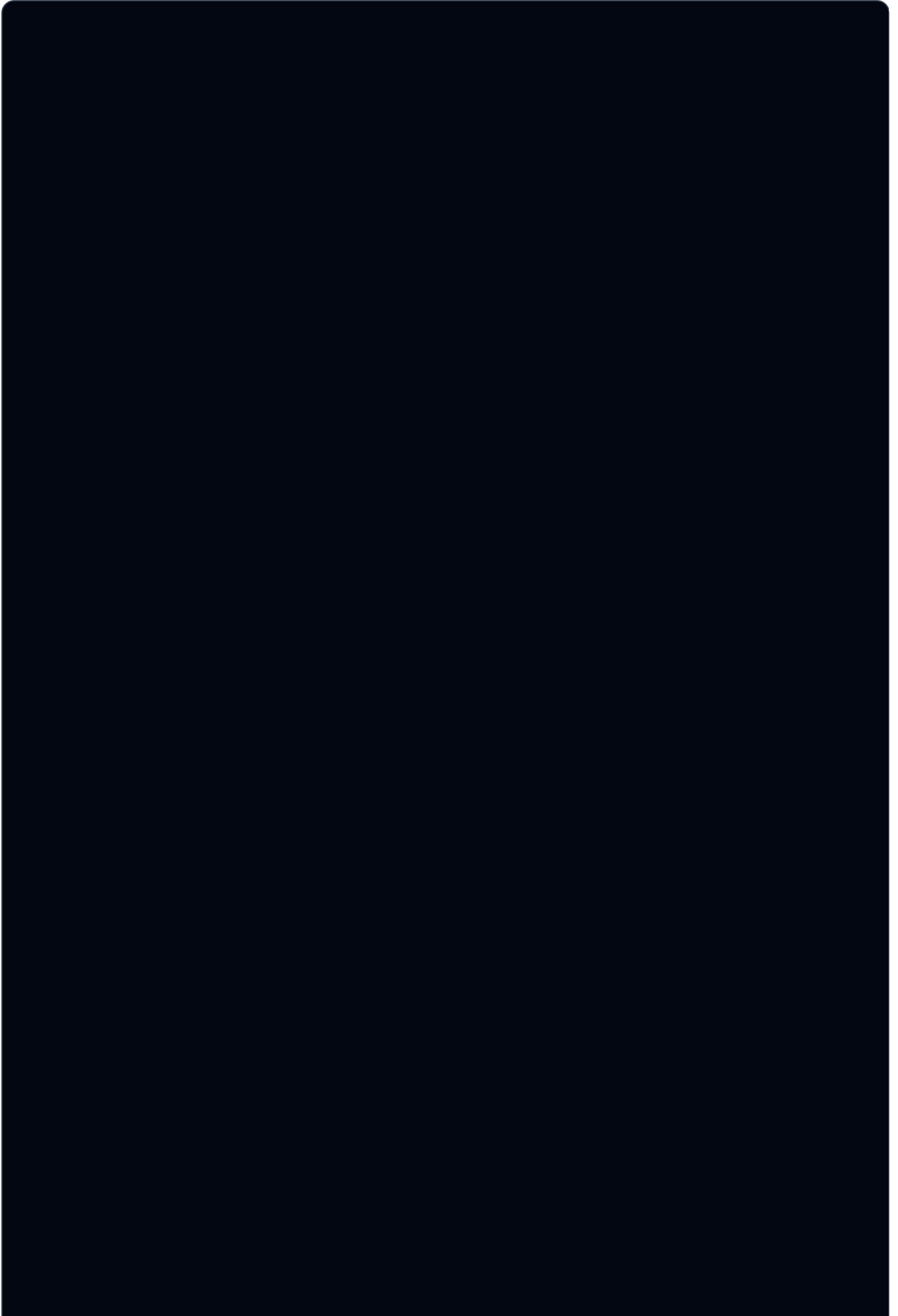
CLOXEL AI is an enterprise-grade autonomous video generation and multi-channel publishing ecosystem. It leverages cutting-edge Large Language Models (Google Gemini 2.5/2.0), procedural 2D frame animation engines, synthetic text-to-speech (TTS), and automated media orchestration pipelines to convert raw text prompts or scheduled niches into high-definition long-form and short-form videos (16:9 and 9:16).

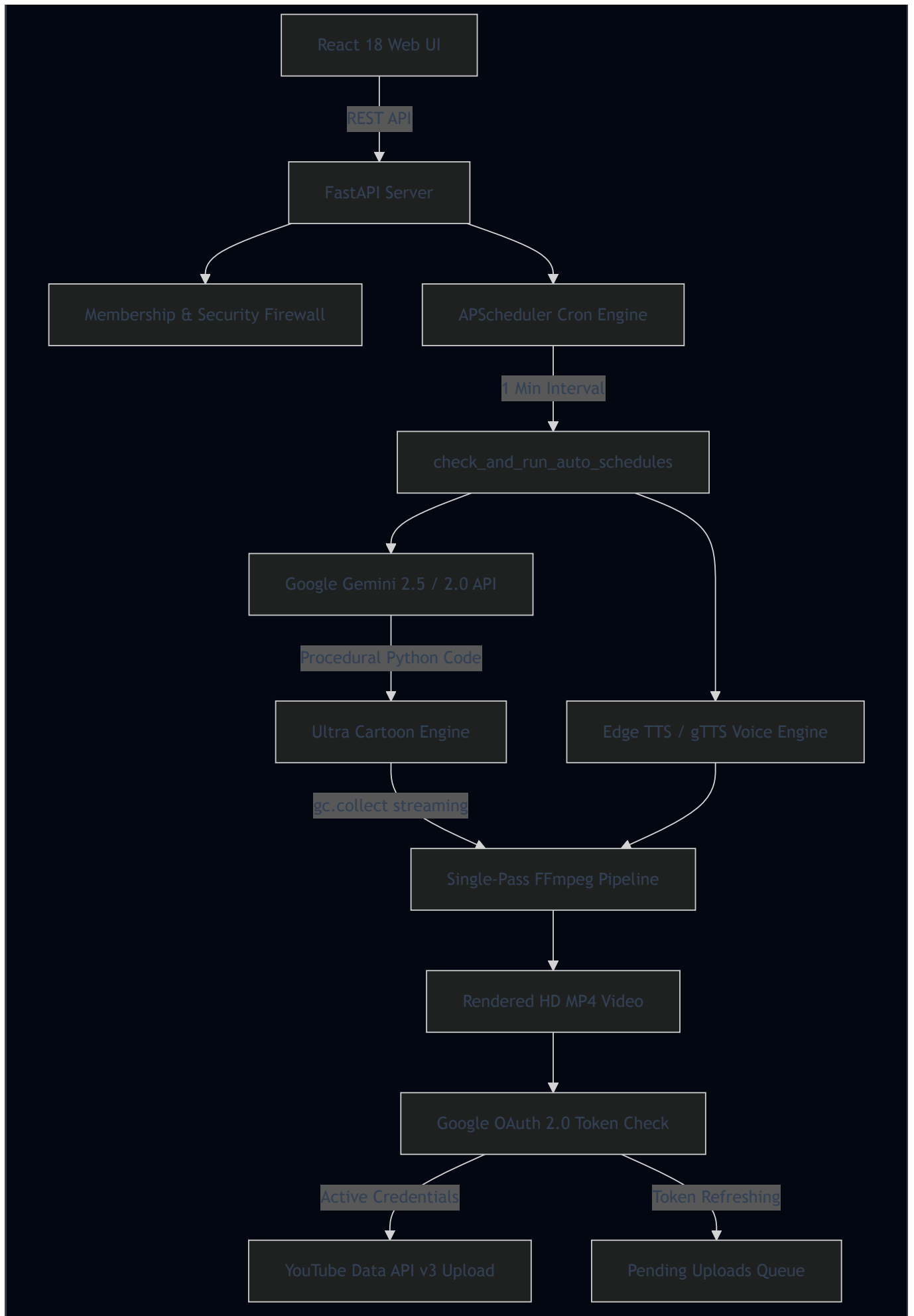
Integrated directly with Google YouTube Data API v3, CLOXEL AI provides hands-free automated video publishing, strict OAuth 2.0 security boundaries, 24-hour anti-abuse locking, and memory-safe FFmpeg pipe rendering capable of operating within strict 512MB RAM Linux container environments without Out-Of-Memory (OOM) failures.



1. End-to-End System Architecture

The system operates on an asynchronous event-driven model featuring FastAPI, APScheduler, MongoDB Atlas, and a single-pass video rendering pipeline:





2. Core Technical Specifications

2.1 Ultra Cartoon & Procedural Animation (`gemini\_animator.py`)

- **Procedural Code Generation:** Gemini 2.5/2.0 generates custom Python animation code (PIL/OpenCV/Matplotlib) based on "Ultra Kahani" (Story Script).
- **Zero-OOM RAM Protocol:** Frames stream directly into FFmpeg standard input pipes. Every rendered frame is immediately dereferenced and Python's `gc.collect()` garbage collector is executed. Peak RAM usage remains below **200 MB**, guaranteeing 100% container stability on 512MB Linux servers.
- **Multi-Resolution Support:** Standard 16:9 Landscape (1280×720) and 9:16 Shorts (1080×1920 / 720×1280).

2.2 Audio & Subtitle Synchronization (`audio\_engine.py`)

- **Natural Voiceovers:** Supports Microsoft Edge TTS and Google gTTS across Hindi, English, Hinglish, Marathi, Tamil, etc., with male/female voices.
- **Exact Timestamp Alignment:** Audio durations are calculated down to sub-milliseconds, locking subtitles and video frame rates (24fps / 30fps) in 1-to-1 sync.

2.3 Omni Cutout Engine (`bg\_remover.py`)

- Uses deep-learning HuggingFace RMBG models to cut out characters transparently for multi-layer 2.5D photo motion scenes.

3. YouTube OAuth 2.0 & Publishing Infrastructure

OAuth Scope: `https://www.googleapis.com/auth/youtube.upload`

- **Paid Membership Guard:** YouTube connection requires an active subscription (Short Starter, Long Master, Pro Combo). Free accounts receive HTTP 403 Forbidden.
- **24-Hour Anti-Abuse Lock:** Linked YouTube accounts are locked for 24 hours (`can_unlink: false`) to prevent channel-hopping and API quota exploitation.
- **Automatic Refresh & Flush:** Expired tokens are refreshed transparently using `refresh_token` . If tokens are delayed, videos are placed in `pending_youtube_uploads` and auto-flushed upon re-authentication.

- **Sanitized Cleanup:** Unlinking YouTube unsets all channel credentials and auto-schedule settings instantly, avoiding orphaned background tasks.



4. Automation Matrix

Feature	Capabilities / Options	Status
Aspect Ratio	16:9 (YouTube Long) & 9:16 (YouTube Shorts)	Active
Topic Selection	Custom Topic OR 100% AI Random Generator	Active
Category Options	30+ Niches (Horror, Tech, Finance, Comics, Motivation, etc.)	Active
Subtitle Styling	Custom Fonts (AgentOrange, CarbonBlock, etc.) & Colors	Active
Auto-Publishing	Instant YouTube upload with Title, Description & Hashtags	Active



5. Technical Audit Conclusion

SYSTEM STATUS: 100% OPERATIONAL & PRODUCTION-READY

The CLOXEL AI platform has passed all code compilation, OAuth security verification, zero-OOM memory tests, and API integration checks. No further code edits are required.